# Analysis Plan - ERP124466

**Study Accession:** ERP124466 (Nov 2018 batch, part of PRJEB40798)
**Study Type:** Global wastewater resistome and antimicrobial resistance surveillance metagenomics
**Sample Type:** Untreated urban sewage (pre-treatment plant influent), frozen and shipped for DNA extraction and shotgun sequencing
**Associated Publications:** PMID 36456547, PMID 38366027, PMID 37430105, PMID 41279477
**Study Design:** Multi-country (101 countries, 243 cities) metagenomic survey of urban sewage to characterize the global resistome and bacteriome. Samples were collected biannually (2016-2021), shipped frozen to DTU Denmark, DNA-extracted with a standardized protocol, and sequenced on Illumina NovaSeq6000 (2x150bp paired-end, targeting >35M read pairs/sample). Downstream analyses include ARG profiling against the PanRes database, microbiome profiling against mOTUs3, metagenomic assembly with metaSPAdes, targeted flanking-region assembly with ARGextender, MAG recovery and within-species phylogenetics, and global distance-decay/network analyses comparing acquired vs. functionally-identified (FG) ARGs.

## Taxonomic / Microbiome Profiling

• **Tool(s):** KMA (with mOTUs3 database); metaSPAdes + MetaBAT2/VAMB for genome binning; GTDB-Tk for taxonomy
• **Database(s):** mOTUs3 (v3.0.3), GTDB-Tk reference taxonomy, Unified Human Gastrointestinal Genome (UHGG) catalog v2.0.2
• **Normalization:** Length-adjusted read counts; centered log-ratio (CLR) and additive log-ratio (ALR) transformations for compositional data
• **Goal of the analysis:** Profile bacterial genus/species composition of sewage, distinguish human-gut-derived vs. environmental taxa, and recover near-complete metagenome-assembled genomes (MAGs) for within-species phylogenetic comparison across regions
• **Key metrics or statistical approaches:** Shannon diversity, PCA/PERMANOVA on CLR-transformed abundances (p=0.001 threshold), Procrustes analysis between bacteriome and resistome ordinations

## ARG / Resistome Profiling

• **Tool(s):** ARGprofiler pipeline (fastp + KMA), ResFinder, ResFinderFG 2.0, Flankophile, ARGextender
• **Database(s):** PanRes v1.0.1 (14,078 unique ARGs combining ResFinder, CARD, MegaRes, AMRFinderPlus, ARGANNOT, ResFinderFG, Daruka et al. functional collection, and MetalResistance)
• **Normalization:** Additive log-ratio (ALR) with bacterial mOTUs reads as reference component; centered log-ratio (CLR) with geometric-mean reference and zero-imputation for compositional statistics
• **Goal of the analysis:** Quantify abundance and diversity of acquired ARGs (ResFinder) vs. functionally-identified (FG) ARGs (ResFinderFG + Daruka et al.); compare regional pan-/core-resistomes and drug-class-level distributions
• **Key metrics or statistical approaches:** Shannon diversity, PERMANOVA (regional beta-diversity, p=0.001), distance-decay linear regression (ln-similarity vs. ln-geodesic distance) at within-country/within-region/between-region scales, Mantel tests (999 permutations)

## Assembly and Contextual Analysis

• **Tool(s):** metaSPAdes (k-mers 27-127, pre-correction flag), ARGextender (KMA + SPAdes recursive targeted assembly), Flankophile, Mash (MinHash sketching), MetaBAT2 and VAMB (genome binning), dRep (dereplication, 95% ANI species threshold)
• **Database(s):** PanRes (for ARG flank targets); PLSDB and PPR-Meta for plasmid classification; Kraken2 Base collection for taxonomic assignment of scaffolds
• **Normalization:** Not applicable (assembly-level)

• **Goal of the analysis:** Reconstruct ARG-carrying contigs and their 1 kb (and 5 kb) flanking regions to determine genomic context, plasmid association, and host taxonomy of the 49 most common ARGs; recover near-complete MAGs for 41 species-level clusters with ≥10 genomes for phylogenetic analysis
• **Key metrics or statistical approaches:** Szymkiewicz-Simpson k-mer dissimilarity (KMA dist) for flank/gene clustering, UPGMA dendrograms with tanglegram entanglement scores, ASTRAL multi-gene species trees, dN/dS via codeml (PAML)

## Geographical / Network Analyses

• **Tool(s):** UMAP (R package umap), correlation-network construction with signed modularity community detection, ggraph backbone layout
• **Database(s):** Natural Earth dataset (geographical shapes for mapping); World Bank regional groupings
• **Normalization:** Hellinger transformation (variant contingency tables); Bray-Curtis dissimilarity (assembled variant resistomes); Euclidean distance on CLR values (Aitchison distance, abundance resistomes)
• **Goal of the analysis:** Determine whether acquired vs. FG ARGs follow distinct geographic dispersal patterns (distance-decay) and identify co-occurring bacteria-ARG community clusters (e.g., the Enterobacteriaceae-dominated FG-ARG community)
• **Key metrics or statistical approaches:** Linear distance-decay models stratified by within-country / within-region / between-region scales with emtrends slope comparisons; network modularity index; Spearman correlation for network edges ($|\rho| \geq 0.5$)

## India-Specific Subset (this curation)

Of the 150 sequencing runs in this batch (PRJEB40798 / ERP124466, November 2018 sampling round), **2 runs originate from India** (sample aliases DTU_2018_838_1_MG_IN_UD_522 and DTU_2018_851_1_MG_IN_KO_535), both satisfying WGS library strategy and bases $\geq 10^9$. These are part of the broader 101-country global sewage collection and were processed identically to all other samples in the dataset (see ARGprofiler methods above). No India-specific reactor/treatment-stage metadata (e.g. temperature, pH, oxygen level) is reported in the associated publications, as this study profiles raw untreated sewage influent only, collected directly upstream of the treatment plant inlet (or discharge point, for sites lacking formal treatment).